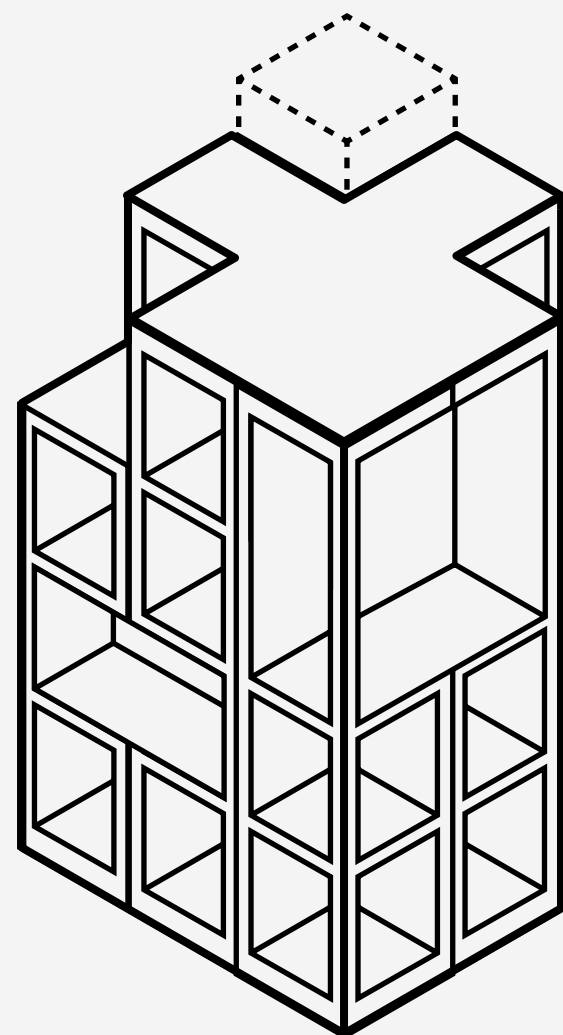


NEXA

Adaptive building platform that lets long-lasting buildings keep up with ever-changing uses

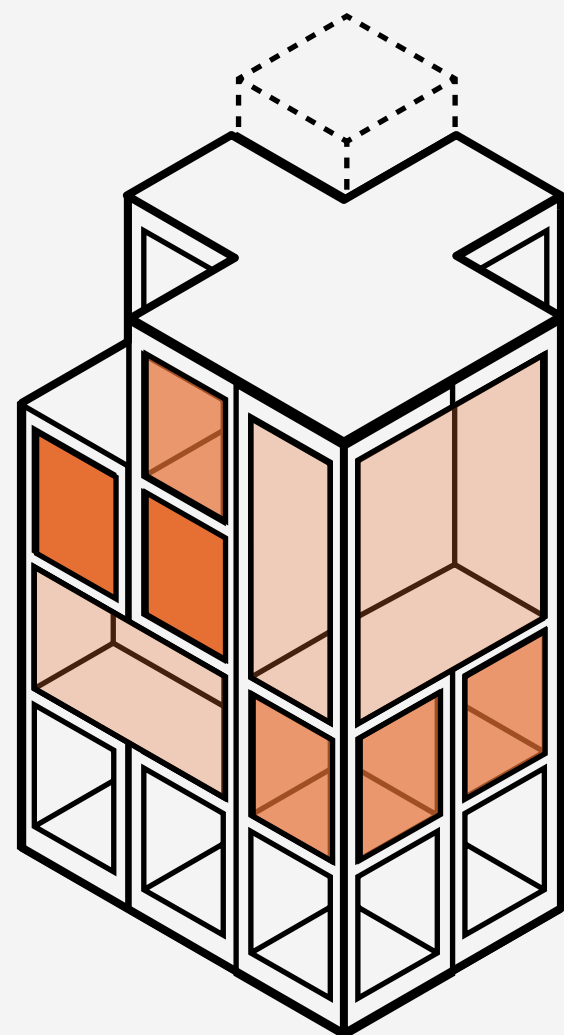
Material Lifespan

Fixed Structural
Frame



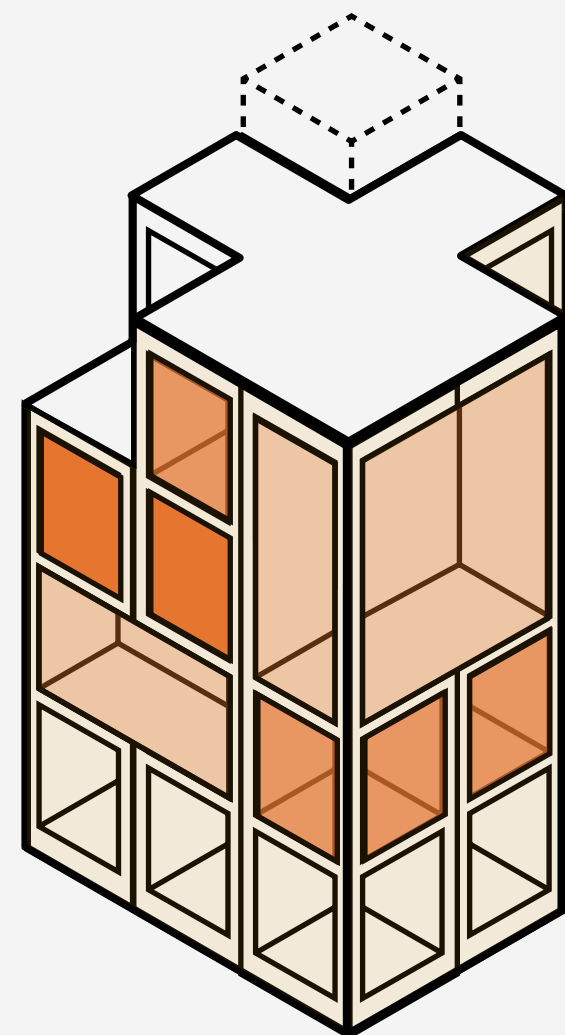
50-100yrs

Infill



30-60yrs

Skin



20-40yrs

Program Timespan

Multi-Unit
Housing

Retail

Food &
Beverage

Refresh
(same user)

15-25 yrs

2-3 yrs

2-3 yrs

Turnover
(new user)

10-30 yrs

3-5 yrs

3-5 yrs

Conversion
(new program)

50+ yrs

7-15 yrs

5-10 yrs

Material Lifespan

Fixed Structural
Frame

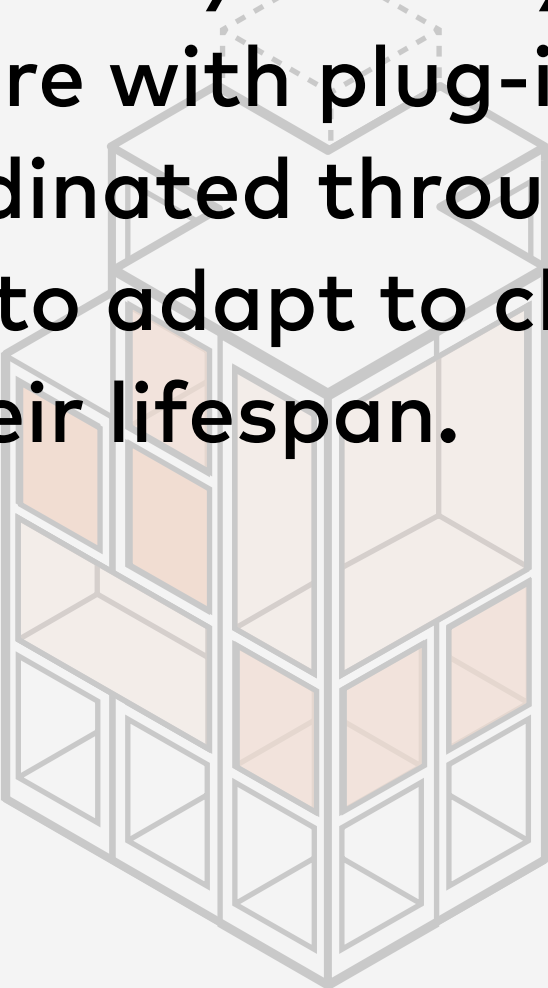
Infill

Skin

NEXA addresses this asymmetry by combining a permanent structure with plug-in, relocatable infill modules coordinated through a digital twin, enabling buildings to adapt to changing needs while extending their lifespan.



50-100yrs



30-60yrs



20-40yrs

Program Timespan

Multi-Unit
Housing

Retail

Food &
Beverage

Refresh
(same user)

15-25 yrs

2-3 yrs

2-3 yrs

Turnover
(new user)

10-30 yrs

3-5 yrs

3-5 yrs

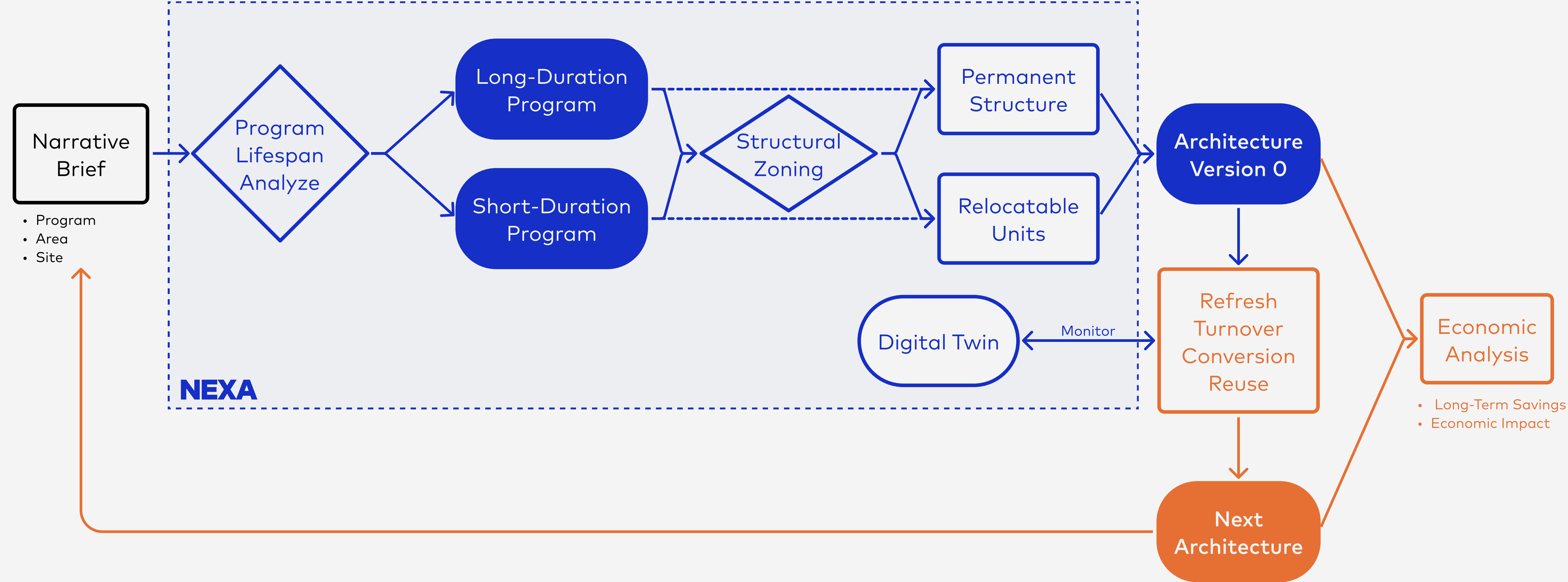
Conversion
(new program)

50+ yrs

7-15 yrs

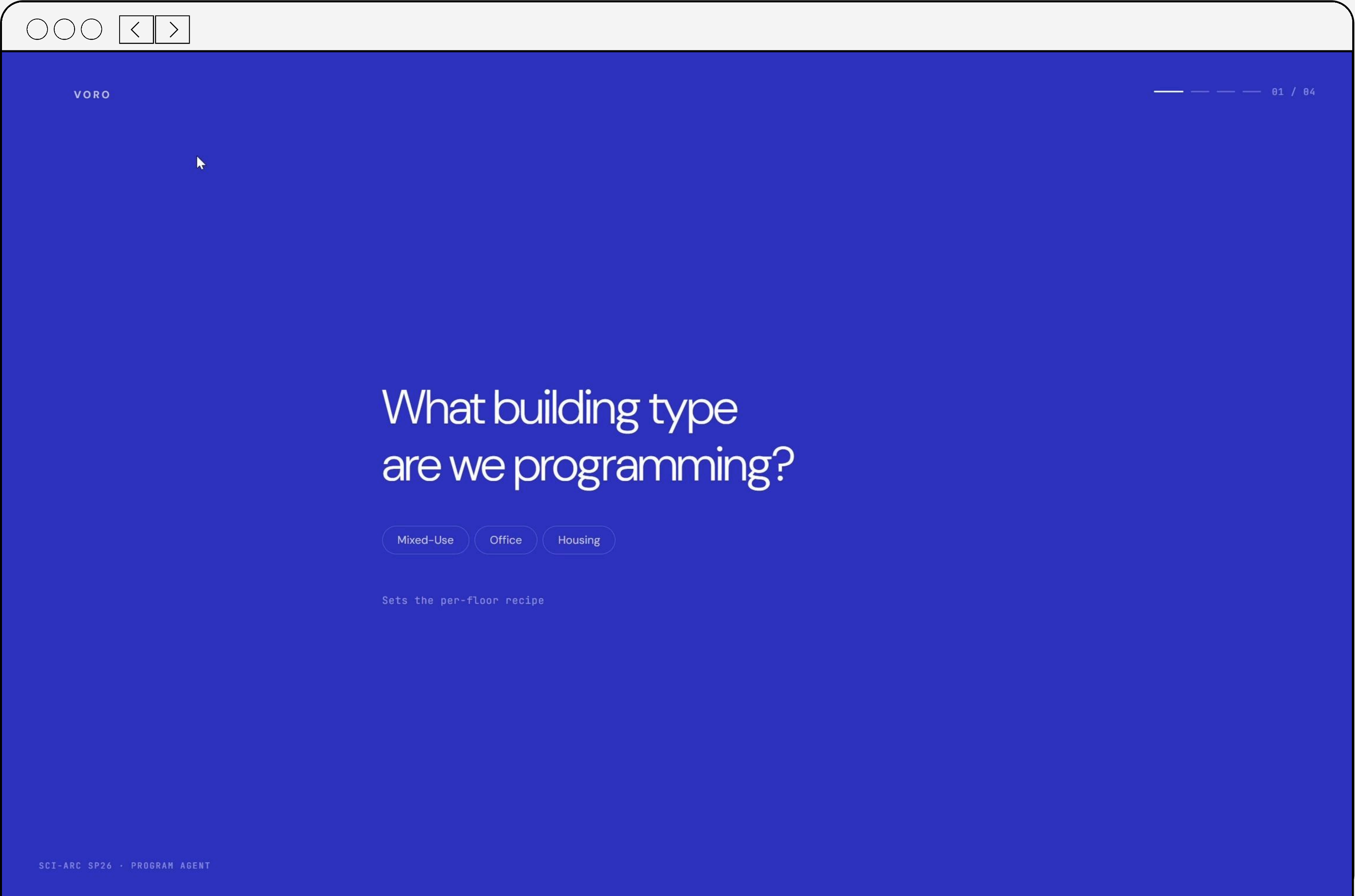
5-10 yrs

From Lifespan Mismatch To Lifecycle Management

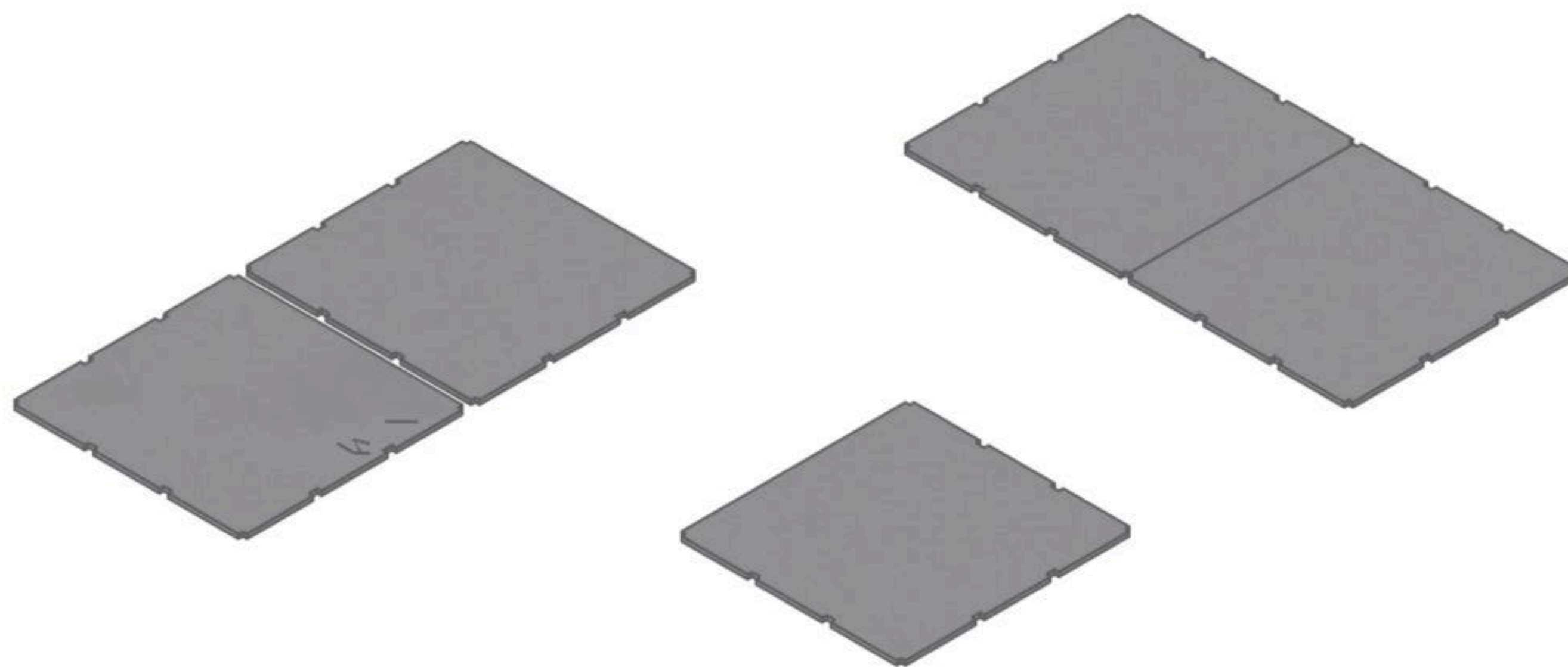


A Platform For How Buildings Evolve

An early-stage, browser-native environmental platform for architects and **developers** to test, simulate and forecast proposals from initial massing through life cycle

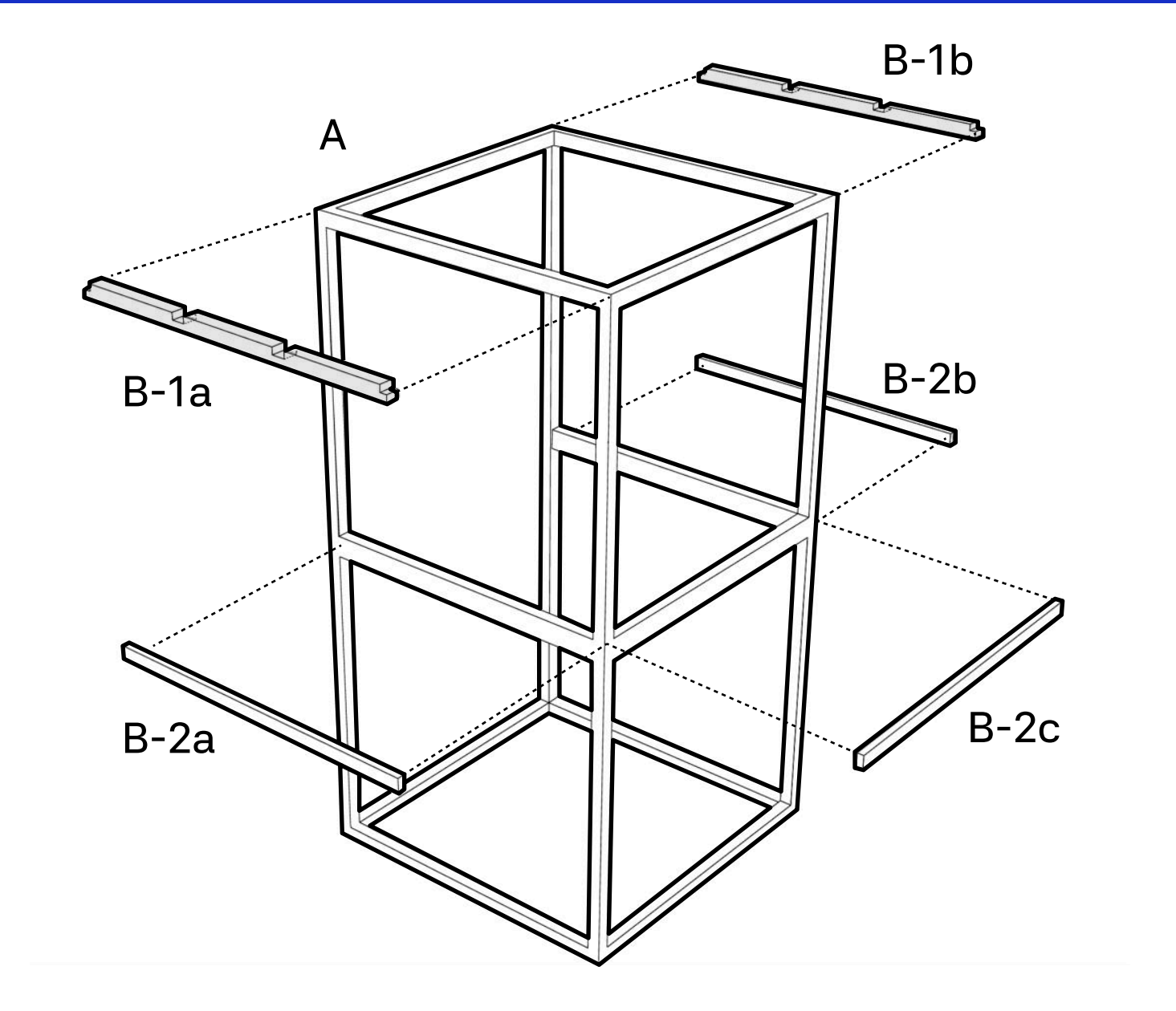


One System, Fully Demountable

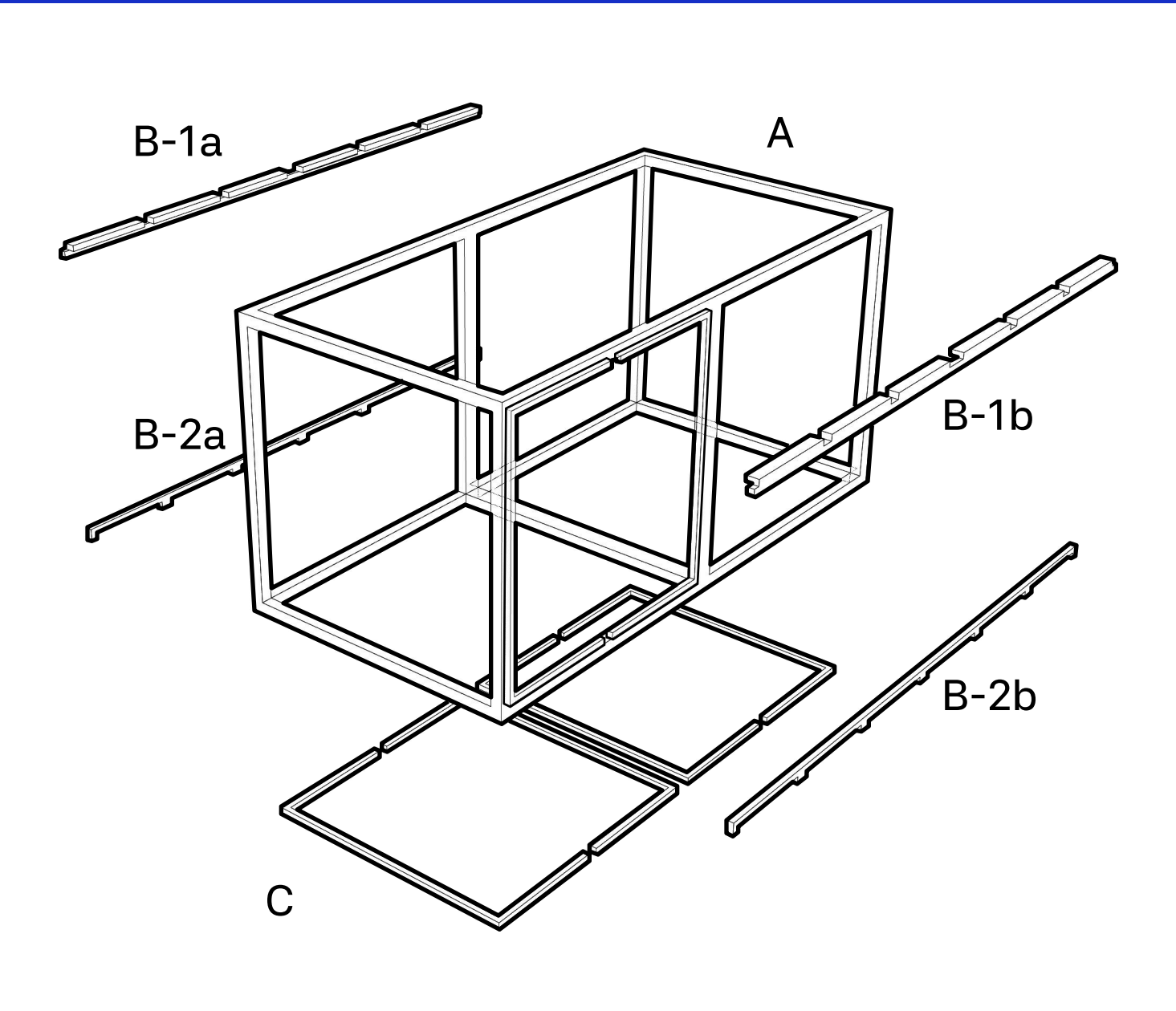


Demountable Frame

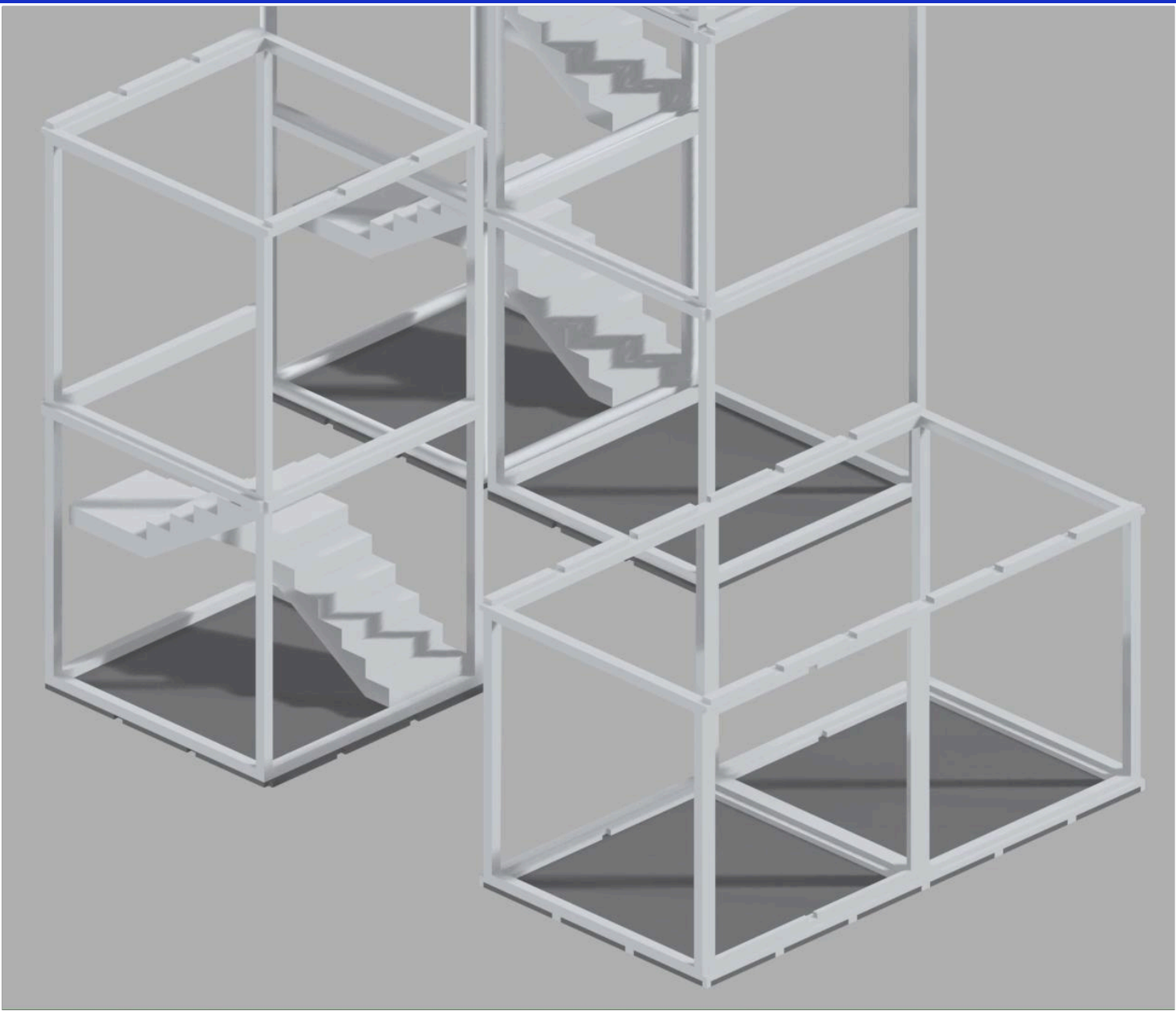
Drop-In → Slide → Lock



Technical specification	FRAME-01	WxL	8'-6" x 8'-6"
		H	17'
		Material	Aluminum

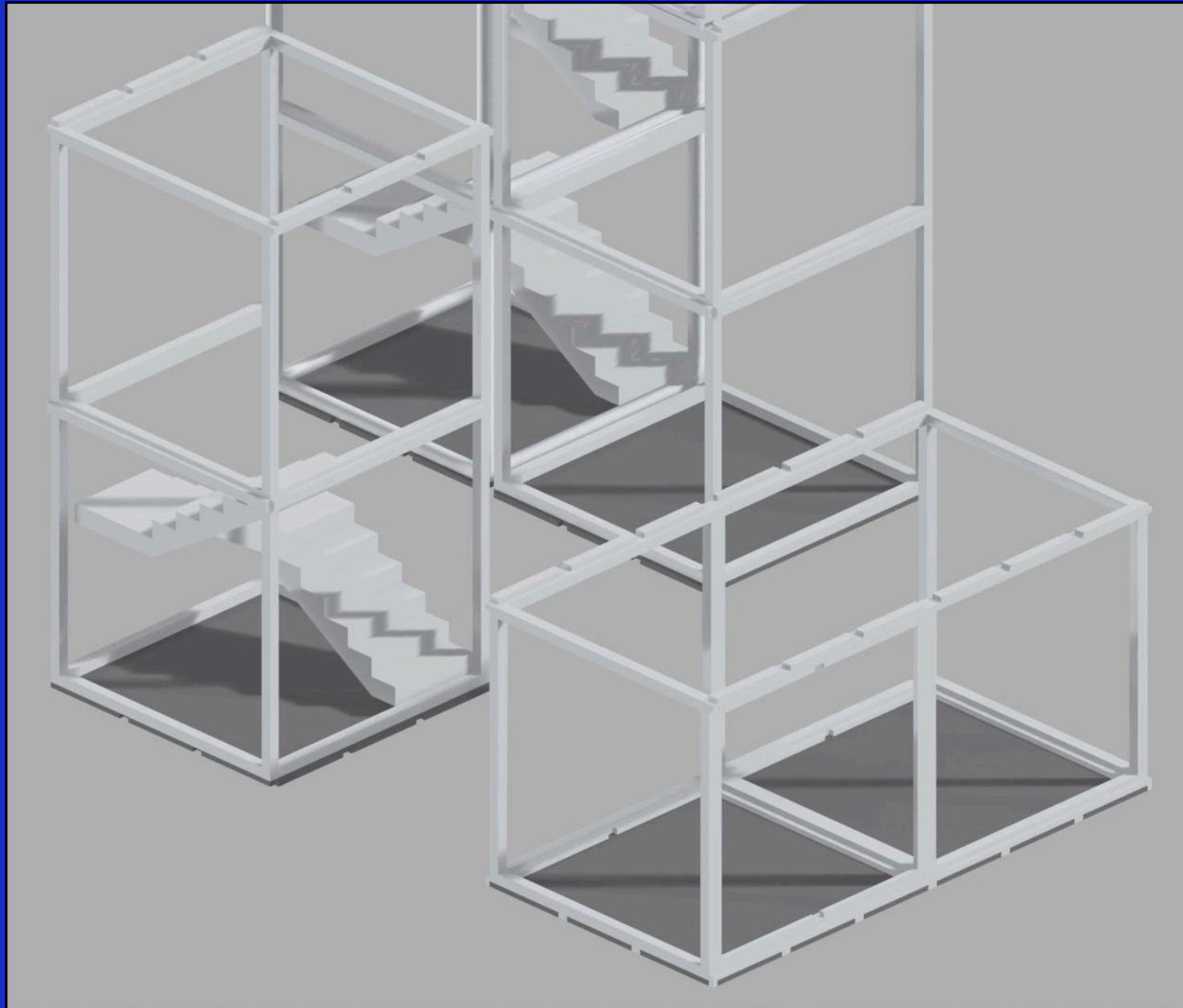
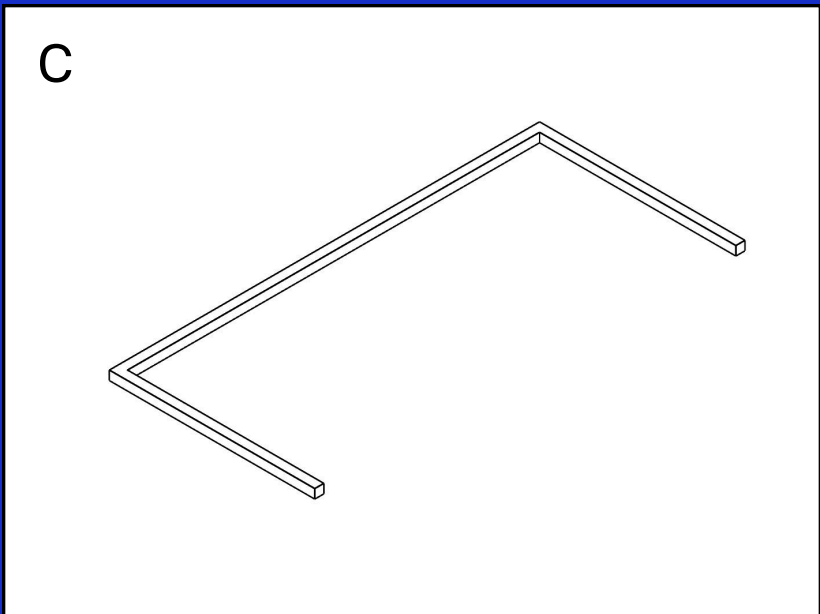
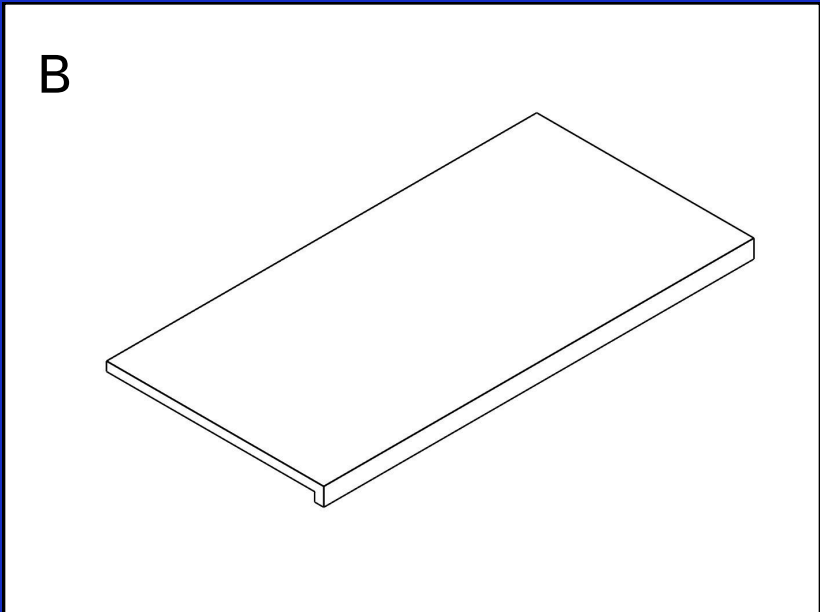
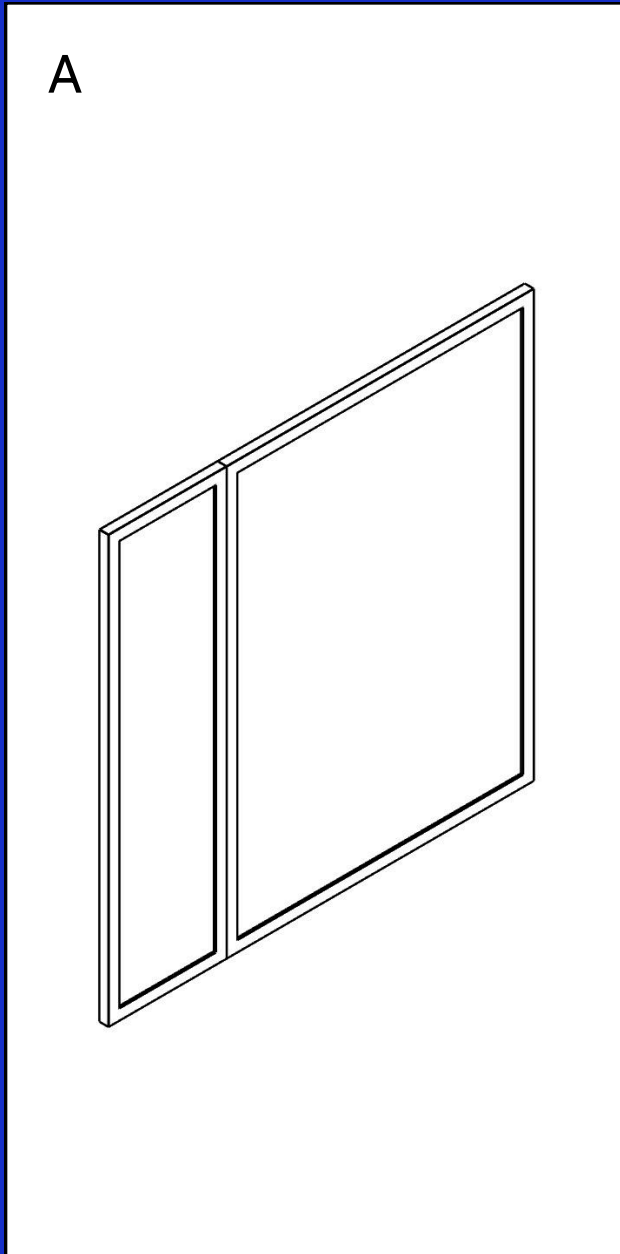
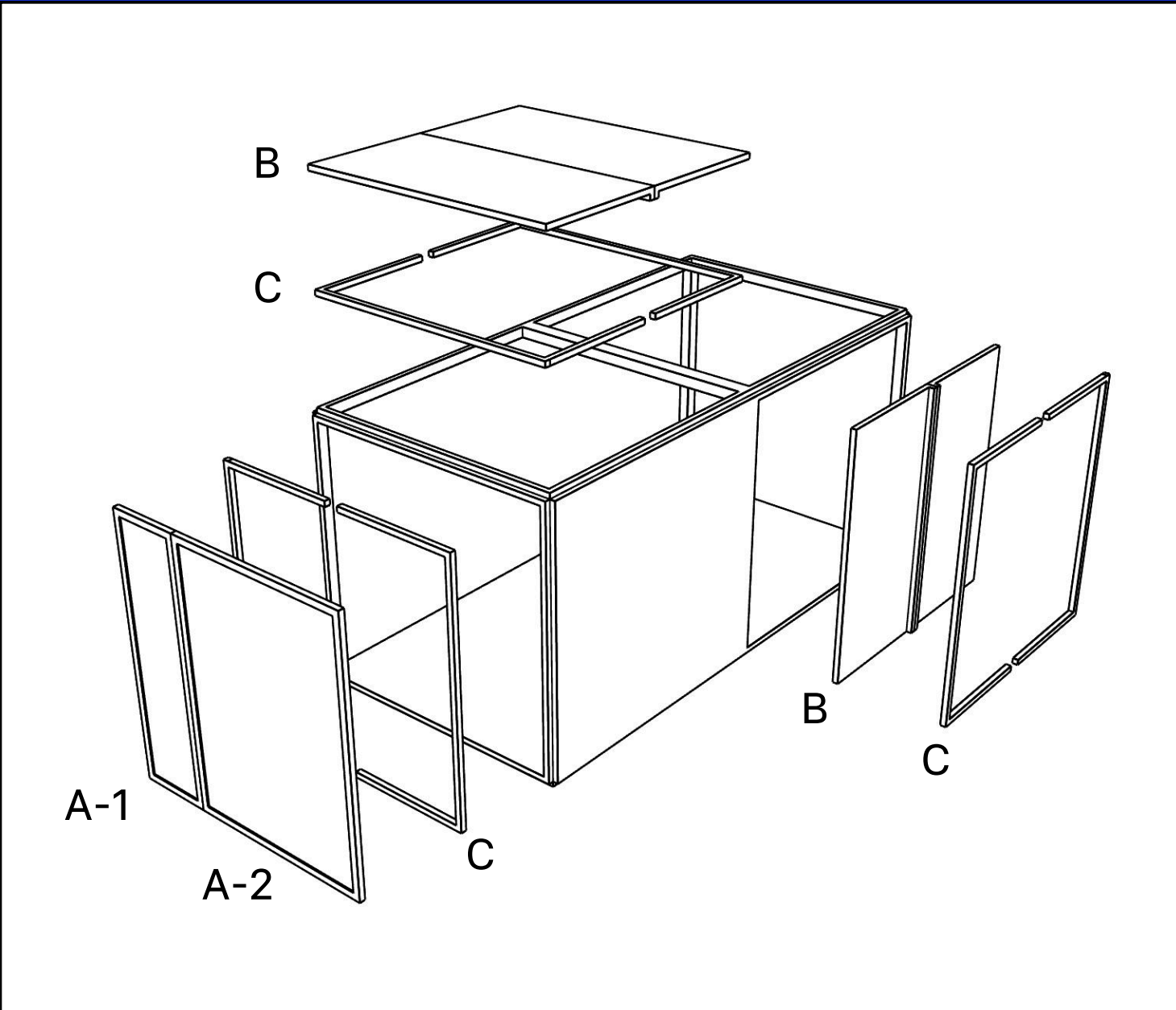


Technical specification	FRAME-02	WxL	8'-6" x 17'
		H	8'-6"
		Material	Aluminum



Technical specification	A	Main frame
	B	Connector
	C	Notched Base Frame

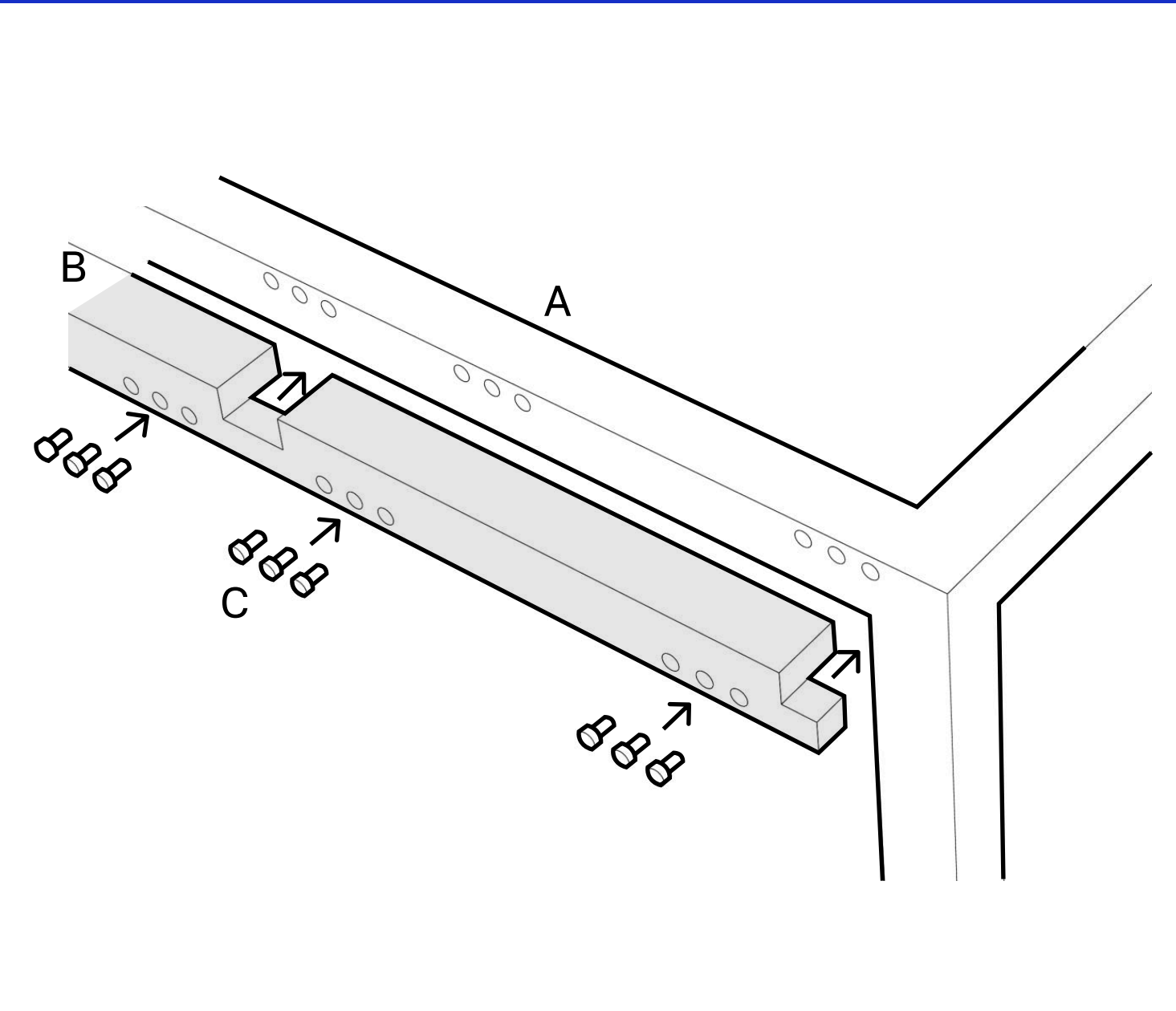
Relocatable Infill



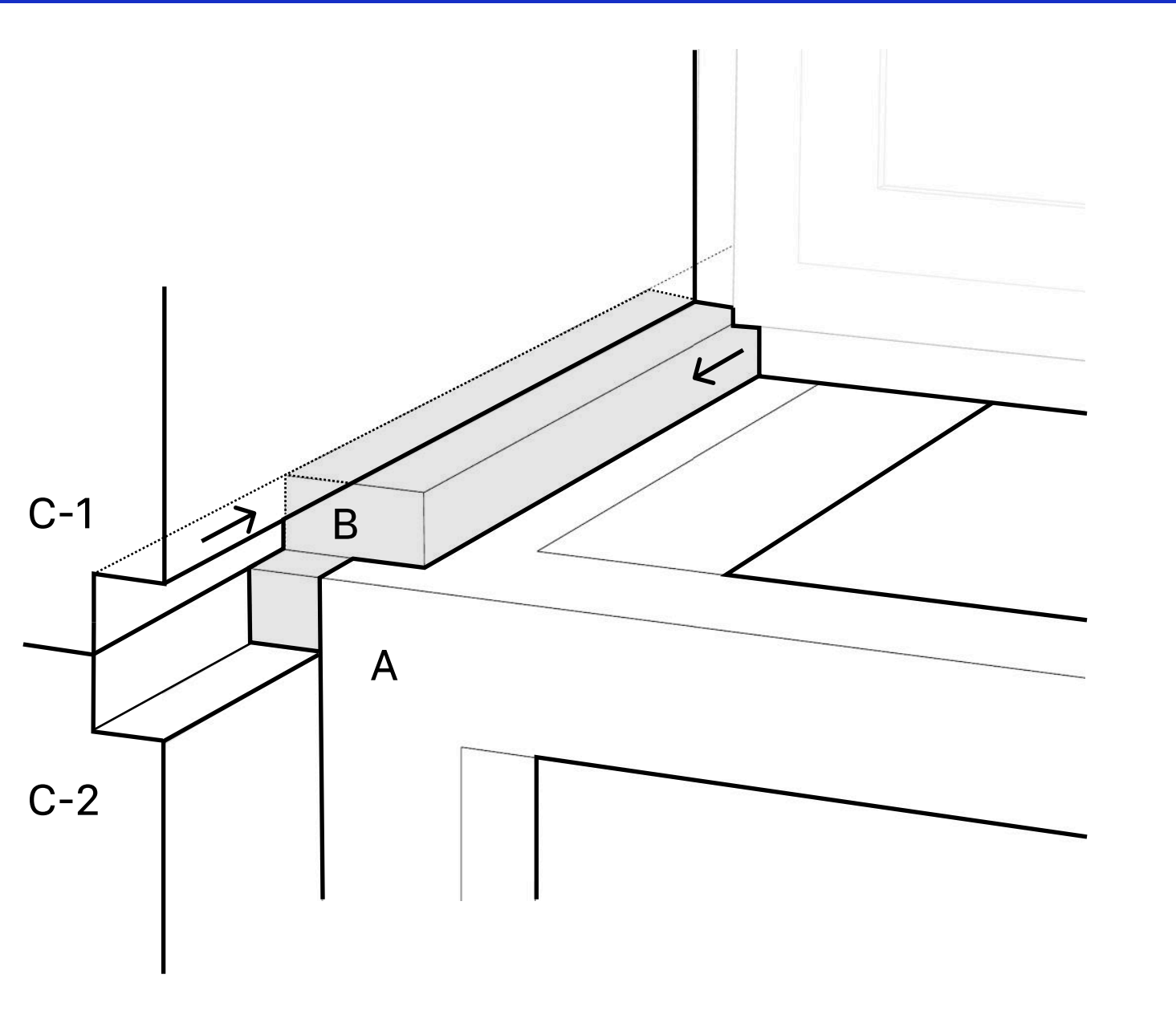
INFILL Technical specification	WxL	8'-6" x 17'
	H	8'-6"
	Material	A

Components	A	Window
Technical specification	B	Notched Deck(floor/wall)
	C	Notched Base Frame

Design-For-Disassembly Connection



Technical specification	A	Main frame
	B	Connector(Notched Rail)
	C	Hidden leveling screws



Technical specification	A	Main frame
	B	Connector(Notched Rail)
	C	Slotted Rail



The Tool The Market Doesn't Have Yet

\$4.2B

Building digital twin market 2026 · 44% CAGR APAC

\$100B+

Global modular construction market 2026

80,000+

US office-to-residential units in conversion pipeline

2–3 yrs vs 50–100 yrs

F&B program lifespan vs. structural frame lifespan

Company	Category	Strength	Gap
Gropyus	Modular Construction	Building OS, robotic prefabrication, standardized housing	Optimizes construction efficiency, not long-term program adaptation
AUAR	AI Modular Design	AI-generated kit-of-parts for customized buildings	Stops after design & construction; no lifecycle management
Madaster	Circular Construction	Material passport, DfD, EU Taxonomy compliance	Records materials but does not generate adaptive building systems
Autodesk Tandem	Digital Twin	Building operation, asset management, facility monitoring	Monitors buildings but cannot reconfigure them
NEXA	Adaptive Building Platform	Design → construction → operation → conversion → reuse in one lifecycle platform	—

No existing platform connects adaptive design, digital twins, and circular construction into one continuous lifecycle system.

~~Building Once~~ → Buildings That Continuously Evolve

01

Adaptive Tectonics

according to program lifespan

Structural permanence and demountable components are divided by the lifecycle of each program — not by cost or construction efficiency alone.

02

Digital Twin—driven Conversion

from monitoring to decision-making

The digital twin is elevated from a passive monitoring tool into an active decision engine that drives spatial reconfiguration and material reuse.

NEXA shifts the design objective from optimizing construction to optimizing building evolution.



Existing platforms optimize how buildings are designed, constructed, or operated. NEXA optimizes how buildings evolve. By using program lifespan as the primary design driver, it integrates adaptive tectonics, design for disassembly, digital twins, and material reuse into a continuous lifecycle platform rather than a one-time construction product.